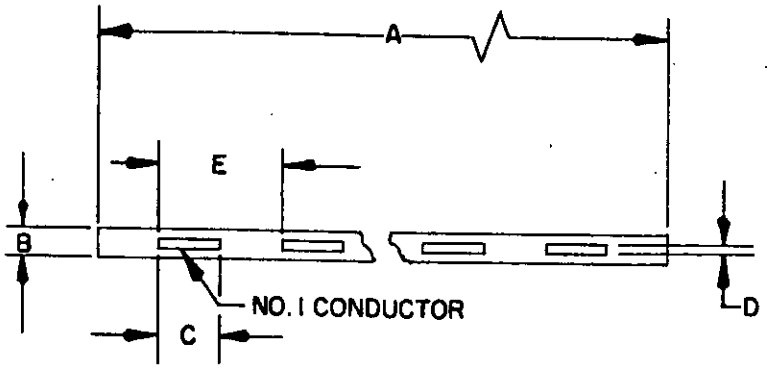
MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL (FLEXIBLE, FLAT, UNSHIELDED) (FLAT CONDUCTOR) 300 VOLTS, 150°C

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cables described herein shall consist of this specification and the latest issue of MIL-C-49059.



INCH	MM	INCH	MM
.0004	.01	.050	1.27
.001	.03	.065	1.65
.002	.05	.100	2.54
.005	.13	.160	4.06
.008	.20	.200	5.08
.015	.38	.5	12.7
.021	.53	1.0	25.4
.025	.64	2.0	50.8

AWG Equiv. (Note 3)	Dimension (inches)				
	A See table I	B (±.001)	C (±.002)	D (±.0004)	E (Note 4)
28	.5, 1.0, 2.0	.015	.025	.005	.050
24	.5, 1.0, 2.0	.015	.065	.005	.100
22	.5, 1.0, 2.0	.021	.065	.008	.100
20	1.0, 2.0	.015	.160	.005	.200
18	1.0, 2.0	.021	.160	.008	.200

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Wire size (AWG) is based on conductor cross-section.
4. Spacing of conductors is typical and non-cumulative.

© FIGURE 1. Flat conductor, flat cable.

© denotes changes

FSC 6145

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Temperature rating: 150°C.

Voltage rating: 300 volts (rms) at sea level.

Conductor coating: See table I.

Cable weight: See table I.

Insulation flaws: Test voltage - AC - 3,500 volts (rms).
Test voltage - DC - 7,500 volts.

Dielectric withstanding voltage: 1,500 volts (rms).

Insulation: Crosslinked modified ethylene tetrafluoroethylene, see table I.

Conductor resistance: See table I.

Accelerated aging and life cycle: Use mandrels coated with polytetrafluoroethylene in the form of either enamel or wrapped tape, such that the diameter of the mandrels, after coating, still conform to dimensions as specified.

Flexing endurance:

AWG equivalent	Temperature rating	Test temperature	Cycles
28 thru 24	150°C	-55°C ±5°C	400
28 thru 24	150°C	150°C ±5°C	400
22 thru 18	150°C	-55°C ±5°C	200
22 thru 18	150°C	150°C ±5°C	100

Part number: M49059/2 - (dash number from table I).

C TABLE I. Dimensions and cable weight. 1/

Dash no. 2/	Conductor size (AWG)	Number of conductors	Conductor coating	"A" Cable width ±.005	DC resistance max ohms/1000 ft	Cable weight max lbs/1000 ft
01	28	10	Bare	0.5	77.0	10.3
03	28	20	Bare	1.0	77.0	20.5
05	28	40	Bare	2.0	77.0	41.0
07	24	5	Bare	0.5	28.1	11.3
09	24	10	Bare	1.0	28.1	22.5
11	24	20	Bare	2.0	28.1	44.9

See footnotes at end of table.

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C TABLE I. Dimensions and cable weight - Continued. 1/

Dash no.	Conductor size (AWG)	Number of conductors	Conductor coating	"A" Cable width ±.005	DC resistance max ohms/1000 ft	Cable weight max lbs/1000 ft
13	22	5	Bare	0.5	17.0	16.8
15	22	10	Bare	1.0	17.0	33.3
17	22	20	Bare	2.0	17.0	66.6
19	20	5	Bare	1.0	11.2	25.3
21	20	10	Bare	2.0	11.2	50.4
23	18	5	Bare	1.0	6.8	37.0
25	18	10	Bare	2.0	6.8	73.9

1/ Dimensions are in inches.
2/ Even dash numbers have been canceled with no military replacement.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, AV, MI
Navy - EC
Air Force - 17, 99
DLA - ES, IS
NSA
MAS
NASA - MSFC

User activities:
Army - AT, ME
Navy - SH

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 6145-0838)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-49059/20	Cable Elec (flex) Flat Cond 300 Volts

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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